

Post-Quantum Cryptography

for the Novice, the Enjoyer, the Deployer and the Academic

Lena Heimberger

BalCCon 2026, Novi Sad

Cryptography is already a part of your life



messages

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messages



payments

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your car

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every connection

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- All of the cryptography is being replaced in the next ten years.
- Touches an **lot** of code.

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- All of the cryptography is being replaced in the next ten years.
- Touches an **lot** of code.
- **Math is too complicated for me**
It's a skillset, not a seperate function.

Mosca's inequality: How long until my secret is revealed?

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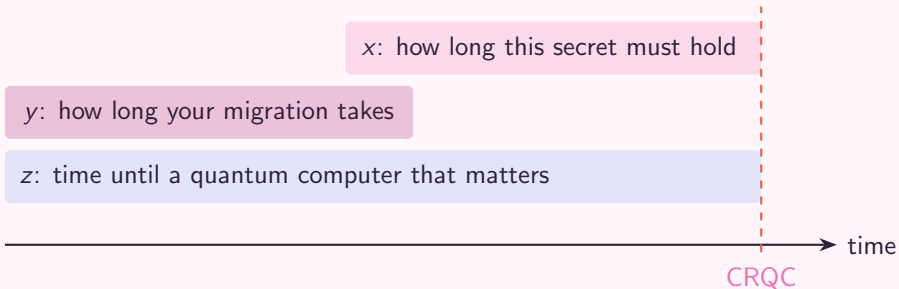
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Mosca's inequality: How long until my secret is revealed?

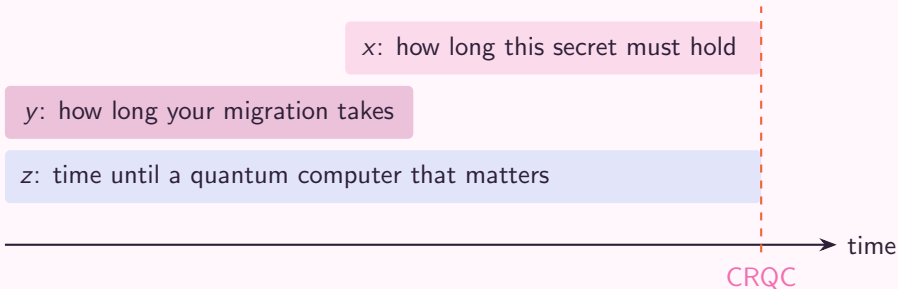
x : how long this secret must hold

y : how long your migration takes

Mosca's inequality: How long until my secret is revealed?



Mosca's inequality: How long until my secret is revealed?



If $x + y > z$, you are already too late.

M. Mosca. *Cybersecurity in an Era with Quantum Computers: Will We Be Ready?* IEEE Security & Privacy 16(5), 2018.

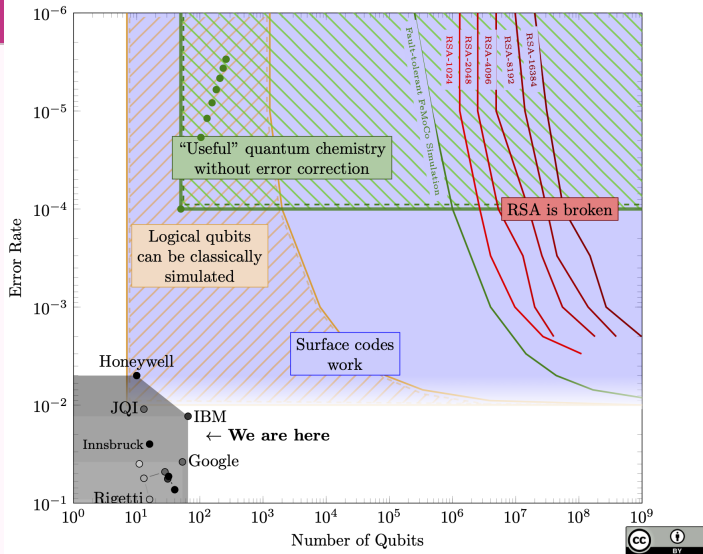


Figure 1: https://sam-jaques.appspot.com/quantum_landscape

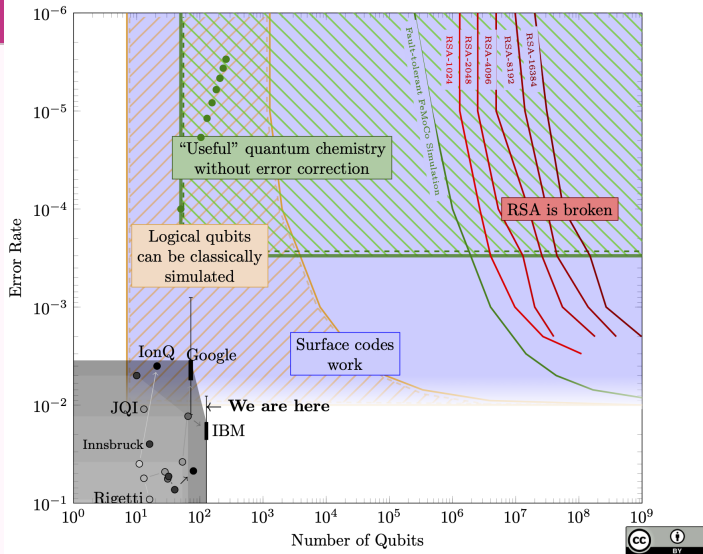


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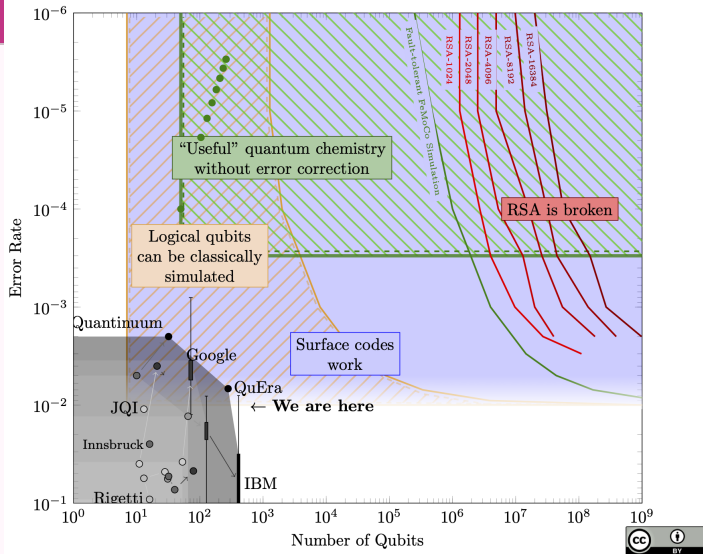


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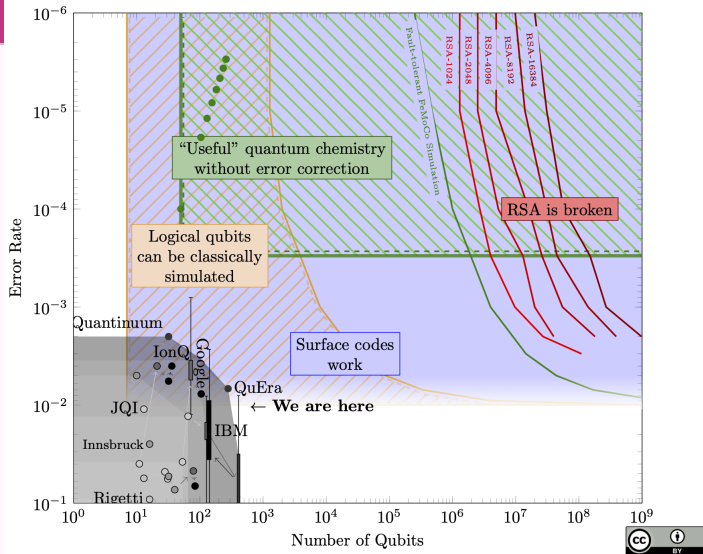


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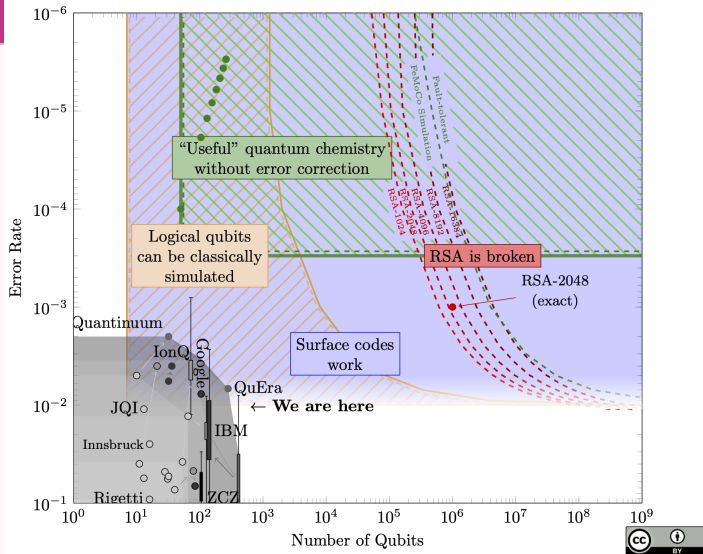


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Where are we?

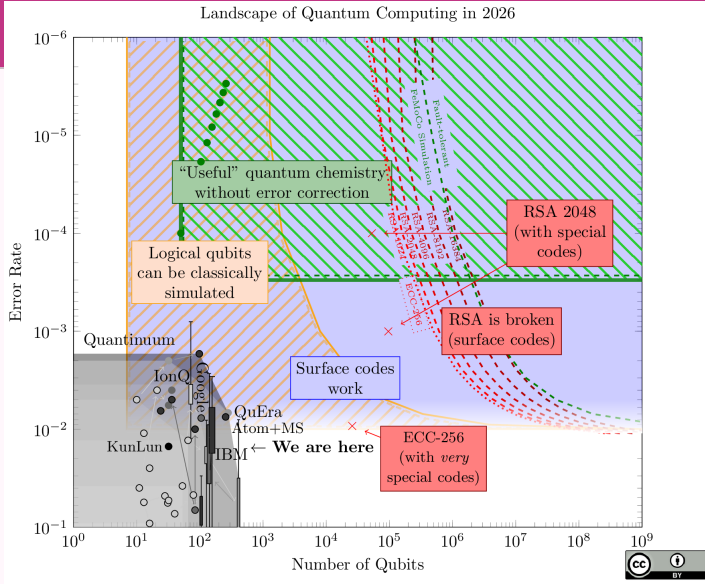
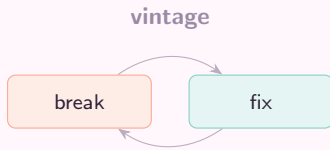
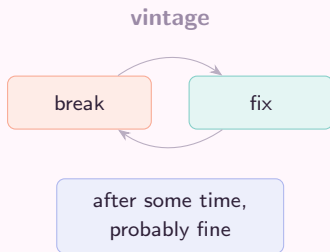


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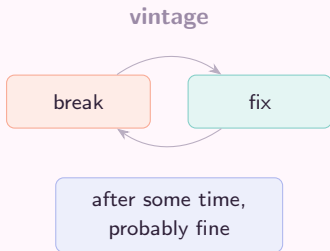
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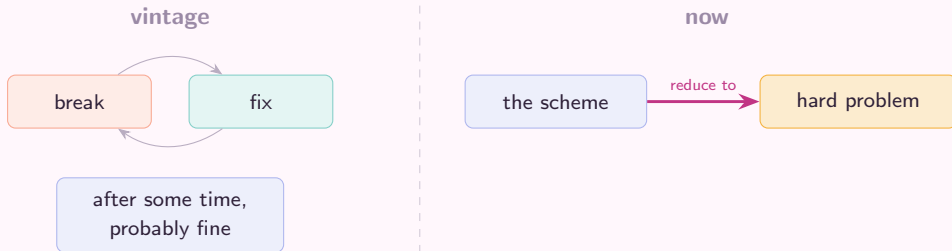


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show breaking the scheme is as hard as breaking underlying assumption

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- Reason about model, tightness of reduction, which problem.

Security Levels

NIST level	at least as hard as
1	recovering an AES-128 key
2	finding a SHA-256 collision
3	recovering an AES-192 key
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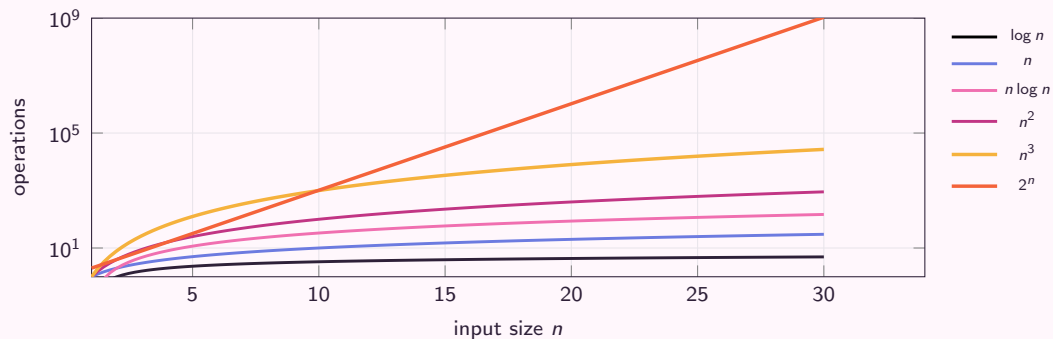
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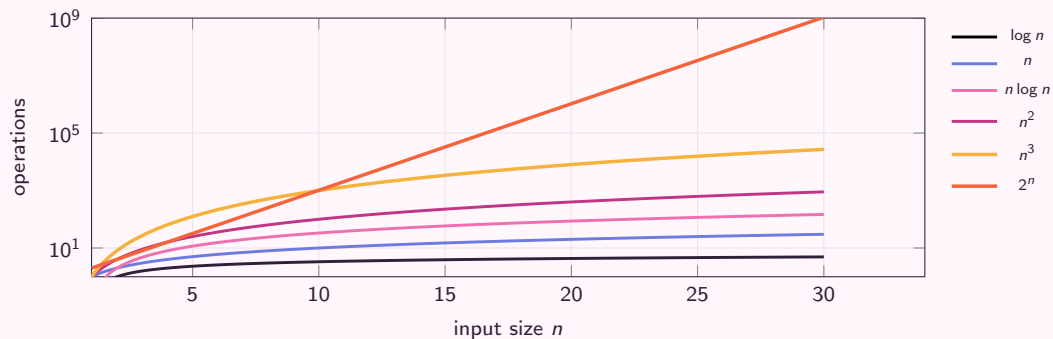
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- Rough estimate, independent of hardware, memory, target...
- **Quantum cost models move:** the qubit count for RSA-2048 reduced by a factor of twenty
How to factor 2048 bit RSA integers with less than a million noisy qubits, Craig Gidney,
<https://arxiv.org/pdf/2505.15917>

Just double the parameters



Also doubles the work you need to do!

Just double the parameters



Also doubles the work you need to do!
Terabytes of RSA keys

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I also did applied crypto last two summers

[Privacy](#) [Terms](#)

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Talk touches a lot of points. If you ever zone out, just zone back in later.

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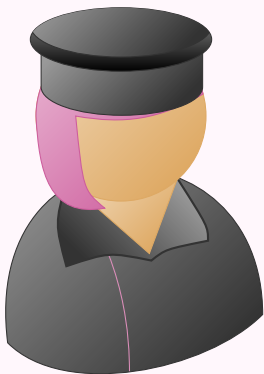
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Signatures

Forgery with a CRQC.

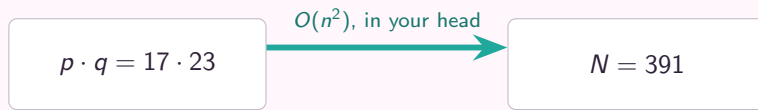
deadline: the day you stop accepting it



the Novice

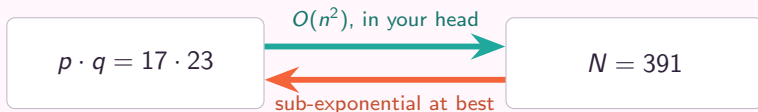
“Why does a quantum computer break cryptography?”

A hard problem becomes easy with a friend's secret



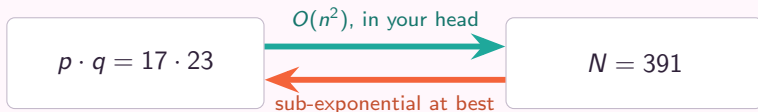
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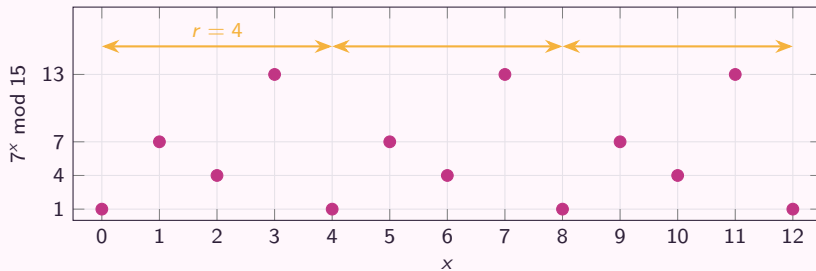


- Polynomial complexity for multiplication.
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- NP-hard: fast verification algorithm, hard to search
bounds the *worst* instance, cryptography needs the *average* instance to be hard.

Shor reduces factoring to frequency finding

factor N

P. Shor, 1994/1997.

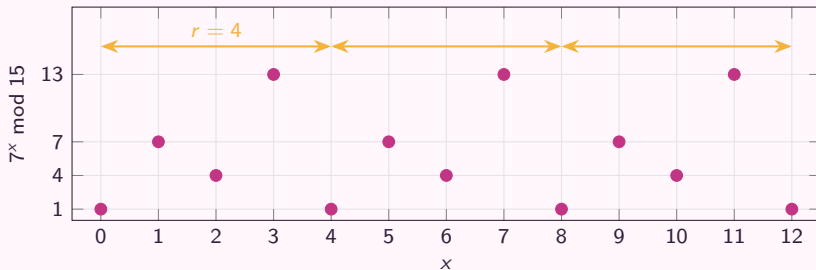


RSA security depends on the attacker not knowing the order of the group.

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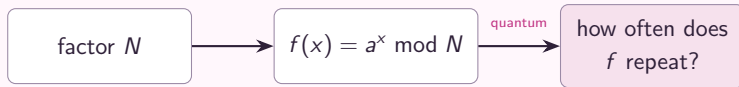


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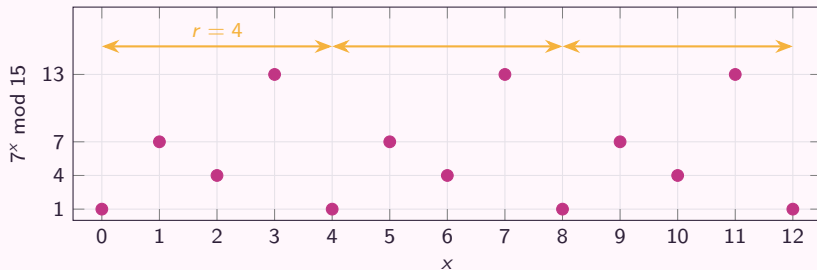


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Blinding operations

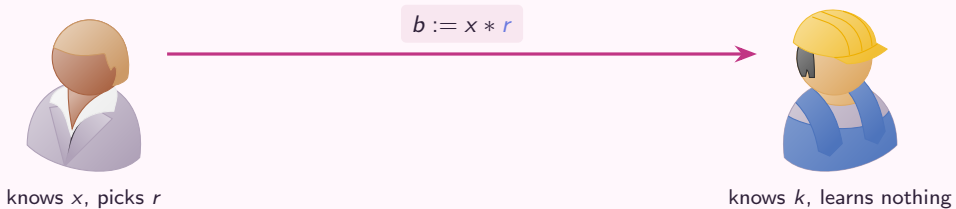


knows x , picks r

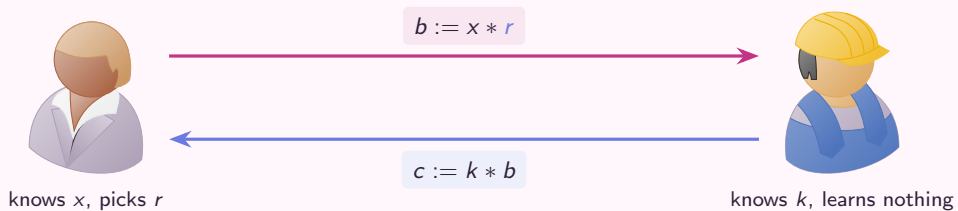


knows k , learns nothing

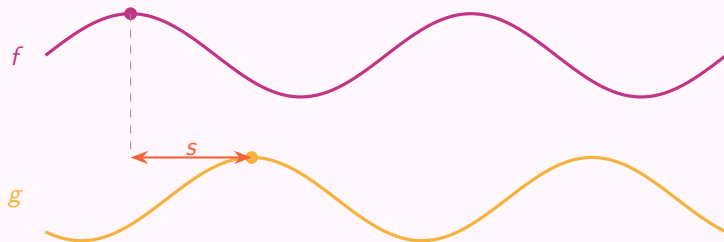
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Blinding operations



Hidden shifts in commutative settings: Kuperberg

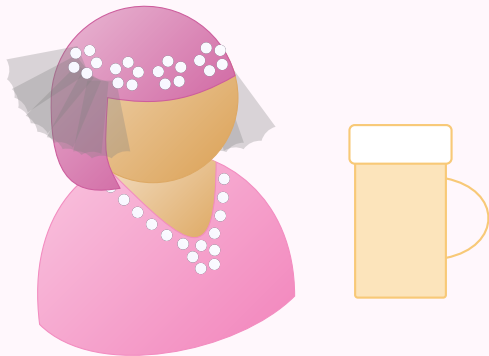


$g(x) = f(x + s)$, and
the secret is now the
shift, not the period

- Same commutative structure, secret moved from *period* to *offset*.
- Then quantum computers have **subexponential** advantage.

G. Kuperberg. A subexponential-time quantum algorithm for the dihedral hidden subgroup problem.
SIAM J. Comput. 35(1), 2005.

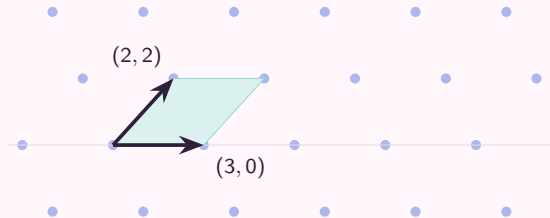
Remove the frequency.



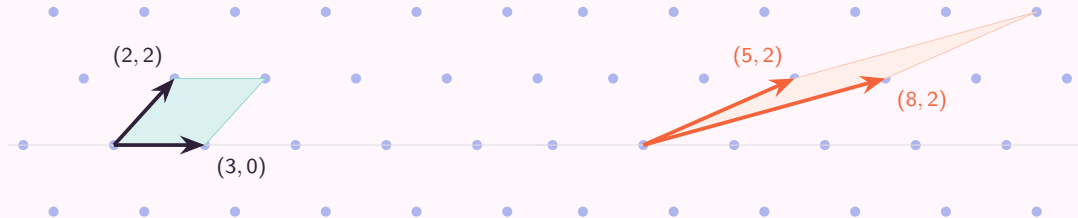
the Enjoyer

“What Alice did last night.”

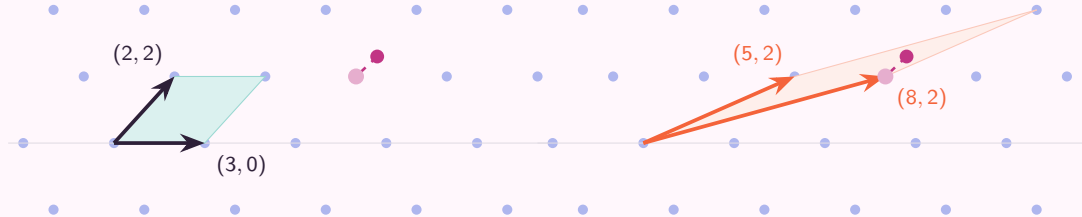
The same lattice, described twice



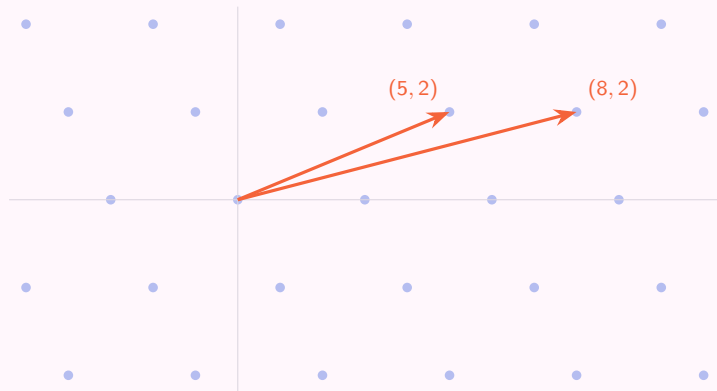
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All about that base



repeat

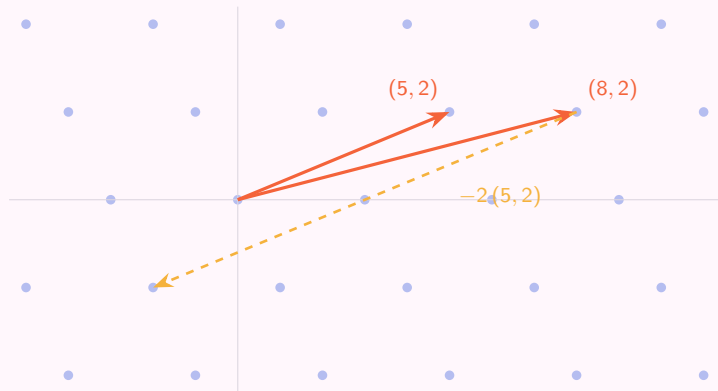
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$$b_2 \leftarrow b_2 - \mu b_1$$

swap if $\|b_2\| < \|b_1\|$

until no swap

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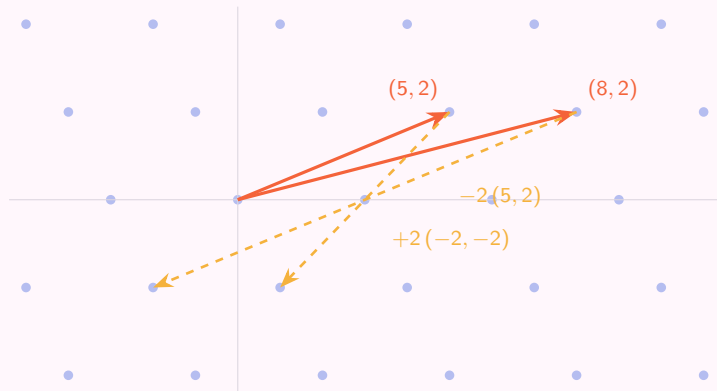
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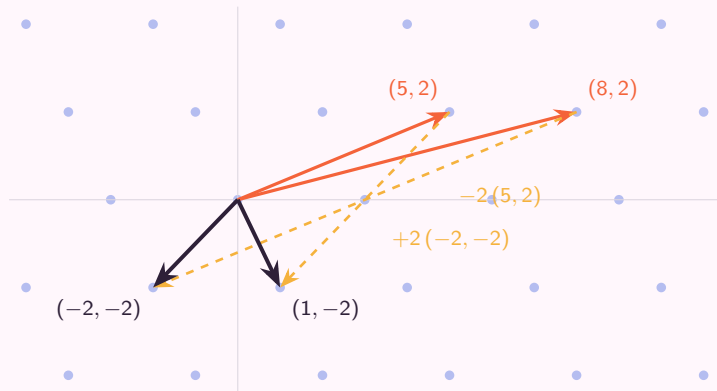
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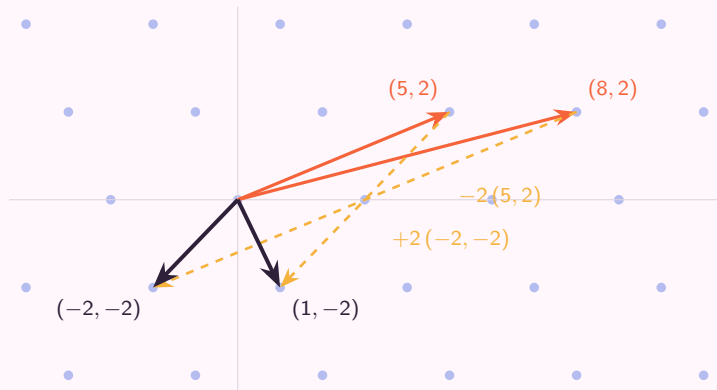
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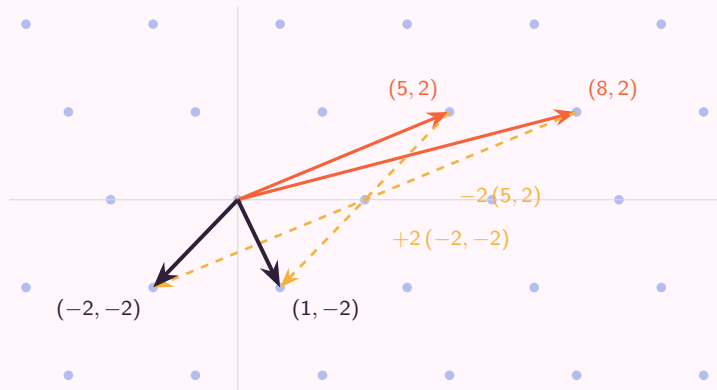
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- LLL is the approximate version.

Lattices: linear algebra with some noise

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Drop e and this is Gaussian elimination.
- with e need lattice reduction
- Everything else is optimization:
Rings and modules to shrink the key, rounding for determinism

A well-studied problem

- system of univariate linear equations over integers with some noise

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Rise from the ashes of cryptanalysis.

Codes: the same equations, counted differently

sent	1	0	1	1	0	0	1	0	1	1	0	1
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two bits flipped

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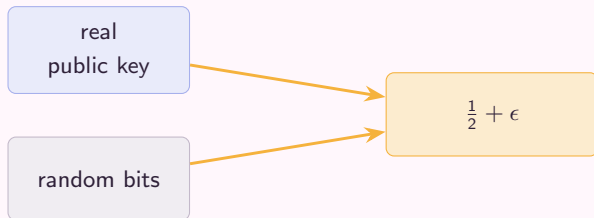
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- conservative choice: Distinguisher attack

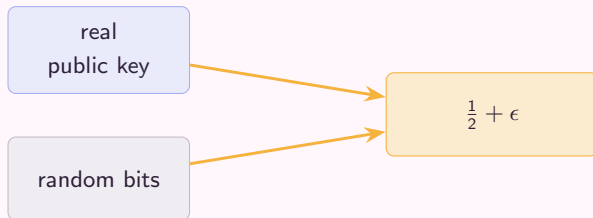
What a distinguisher actually means



Quasipolynomial Cryptanalysis of the McEliece Cryptosystem (or: PIR Meets McEliece)

Ghoshal, Ishai, Jain, Sun, ia.cr/2026/1630

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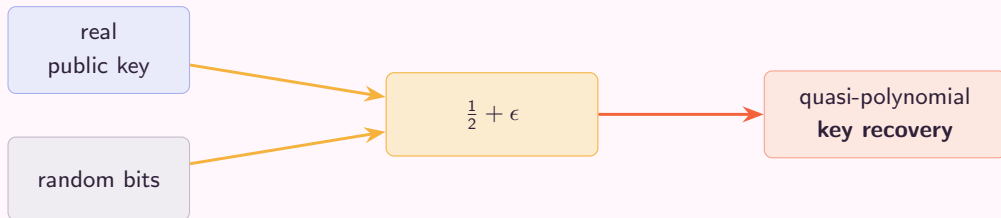


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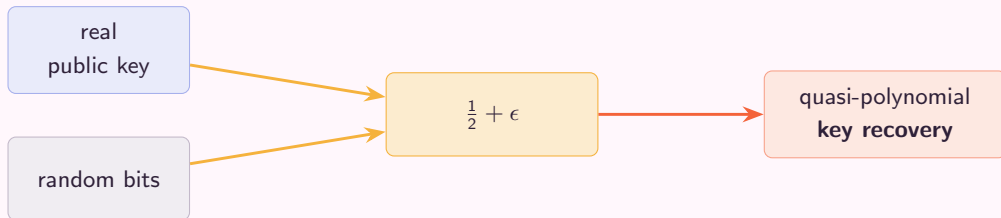


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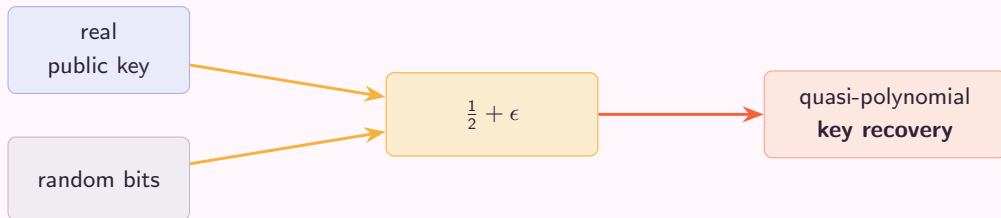


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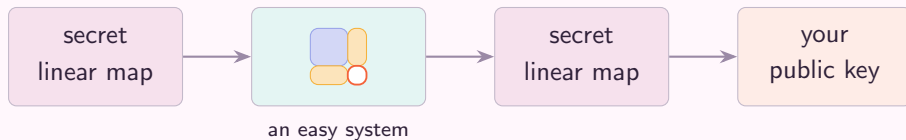


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- distinguisher became a key recovery in a few weeks
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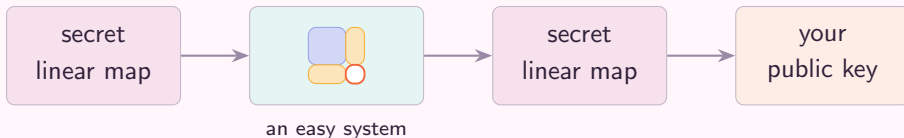
Quasipolynomial Cryptanalysis of the McEliece Cryptosystem (or: PIR Meets McEliece)

Ghoshal, Ishai, Jain, Sun, ia.cr/2026/1630

Multivariate: hide an easy system inside a hard-looking one

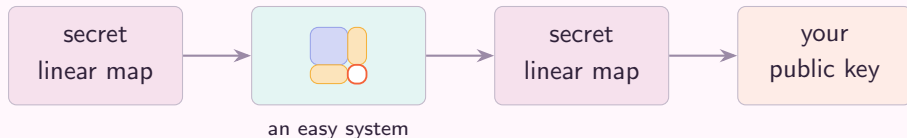


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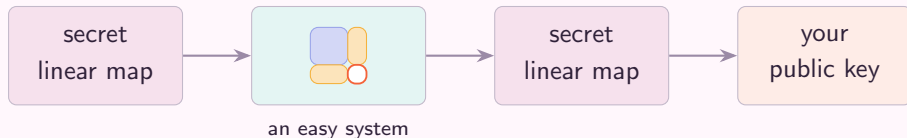
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- **No noise anywhere.** Every equation holds exactly. The only thing hiding the structure is the two linear maps, so attacks aim straight at them.
- 96-byte signatures, public keys in the tens of kilobytes, and a habit of dying suddenly.

The easy system: oil and vinegar

vinegar	oil
$V \times V$	$V \times O$
$O \times V$	0

no oil times oil: fix the
vinegar and the rest is
linear

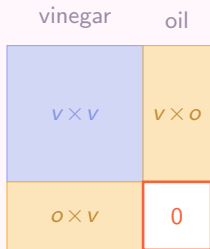


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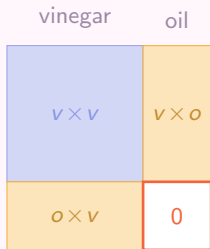
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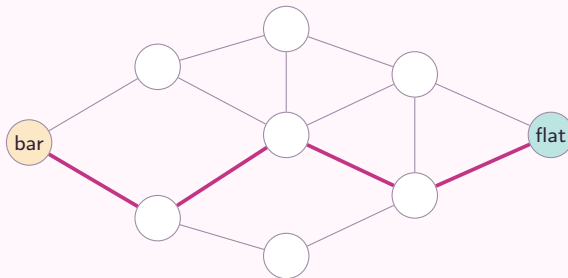
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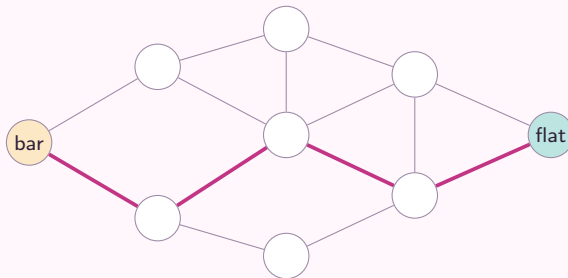
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- **Balanced** oil and vinegar died in 1998 (Kipnis-Shamir). **Unbalanced**, two or three times as much vinegar, is still standing after twenty-five years.
- Every attempt to shrink that public key with more structure has been risky. Rainbow, a layered variant, fell in a weekend on a laptop in 2022.

Alice walks home alone at night



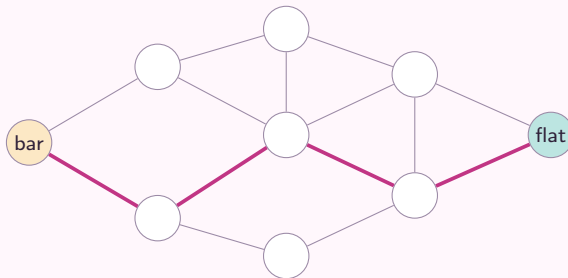
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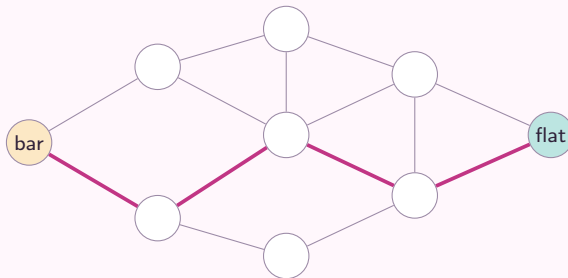
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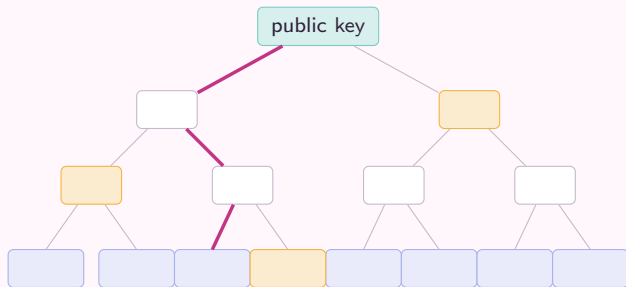
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- SIDH, the previous favourite, was broken in an afternoon in 2022.

You young people with your fancy math!
We know how to build cryptogrphahy

Symmetric: One way to signatures



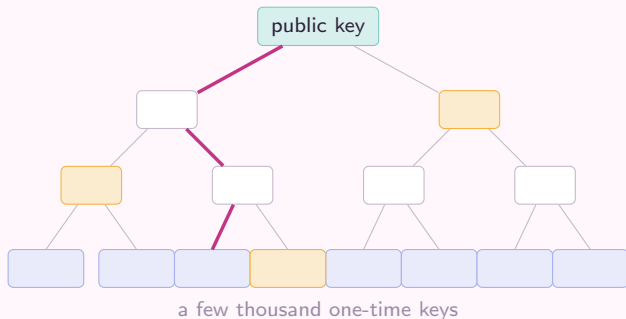
a few thousand one-time keys

siblings: the rest of the path

the signature is this path

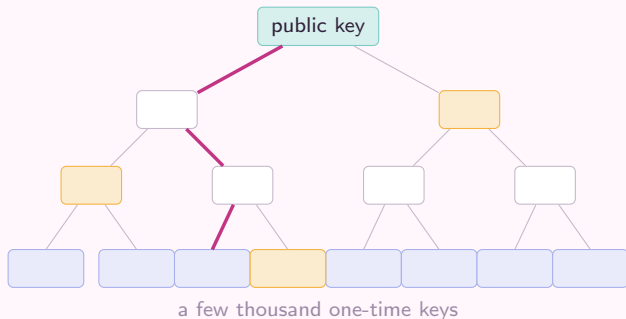
one one-time key, used once

Symmetric: One way to signatures



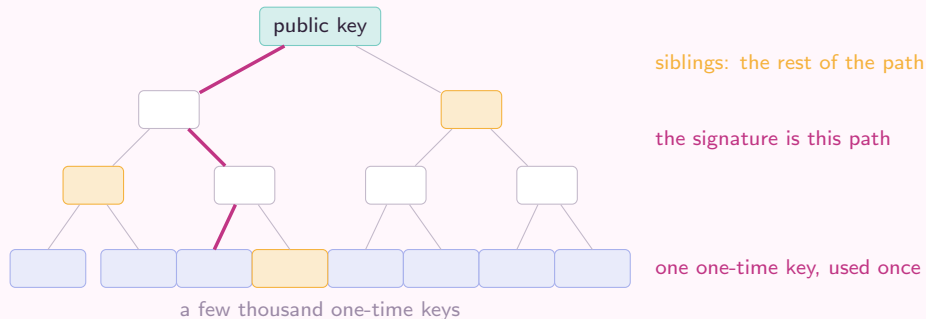
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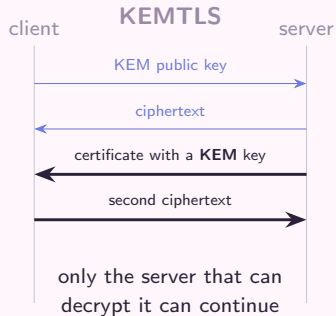
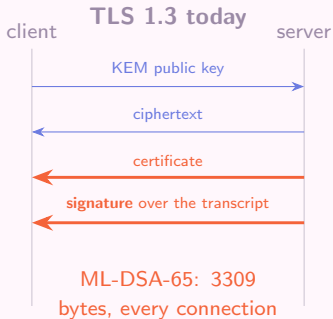
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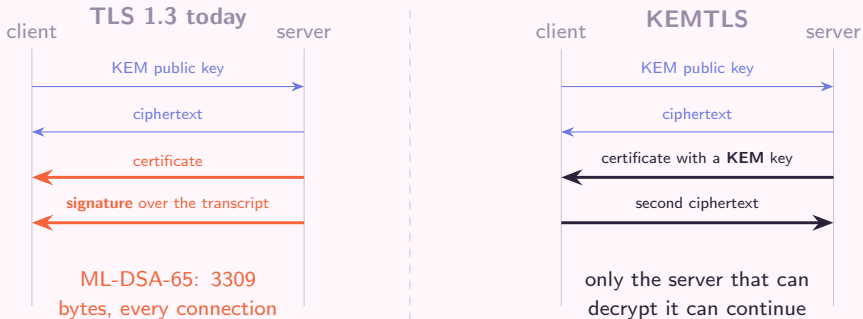
- A hash turns a secret into a one-time key.
- Clunky: either keep state and never reuse a leaf, or go stateless and large
- Security reduces to underlying hash function. We need one anyways, somewhere

But signatures are expensive!

KEMTLS: authenticate via decryption

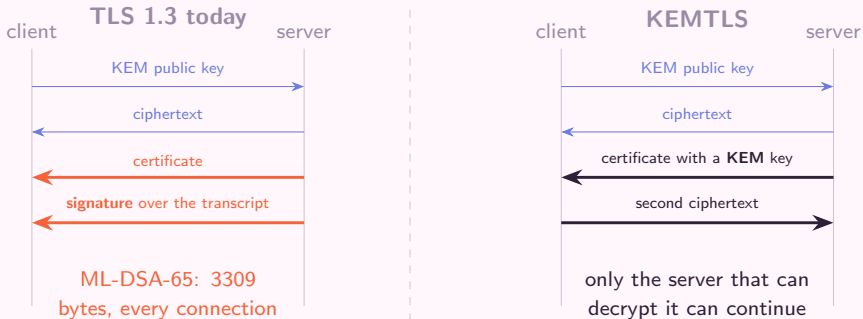


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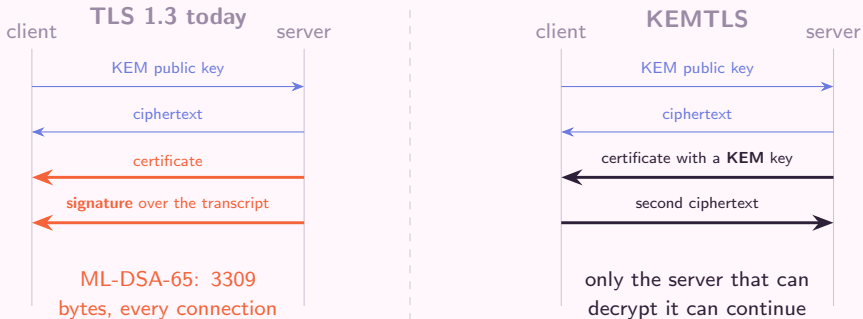
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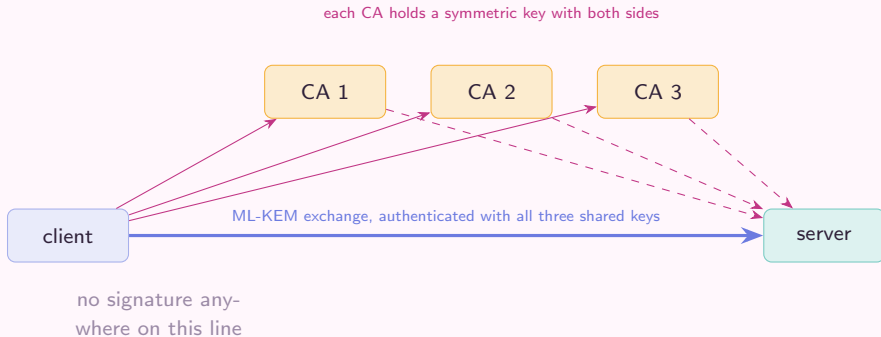
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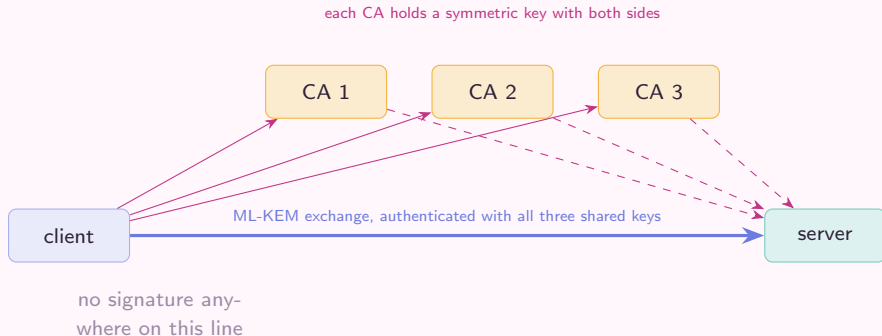


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- Price: authentication is **implicit** at first, and the server's first data waits one extra flight.
- The CA still signs your certificate. KEMTLS removes the one signature made live; Merkle tree certificates remove the four made in advance.

The escape hatch nobody wants: symmetric all the way down

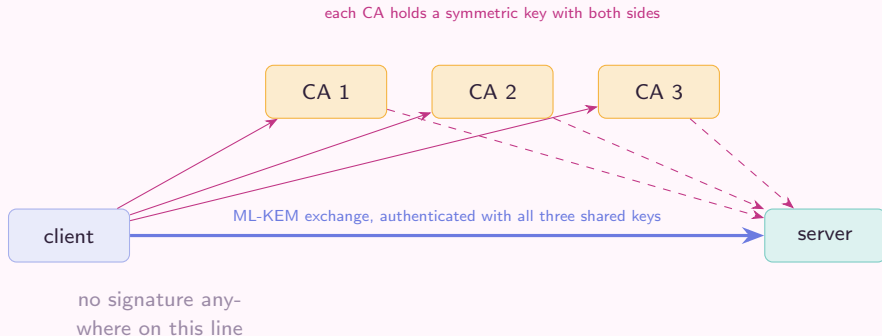


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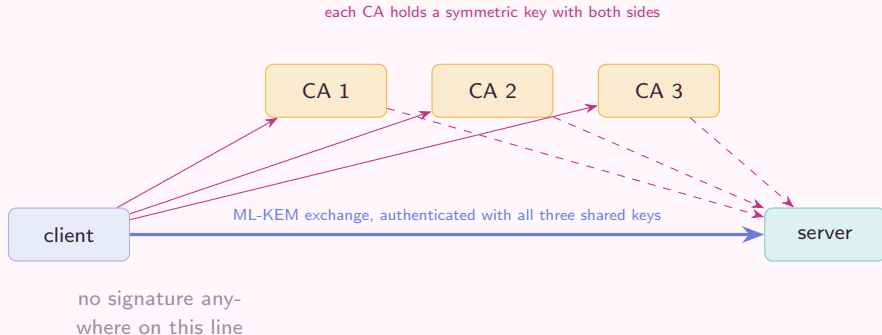
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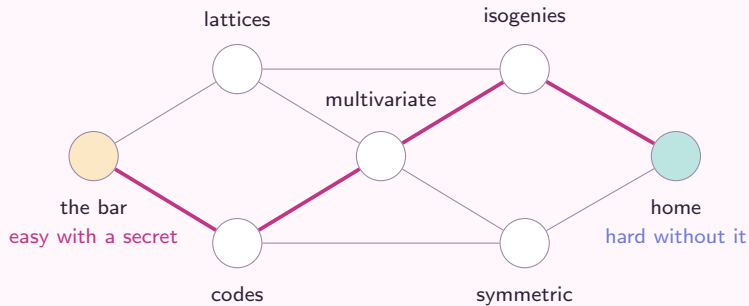
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- **Three CAs must collude** for an attack.
- every CA learns which sites you visit, live. Like OCSP, but worse.

Adam Langley, <https://www.imperialviolet.org/2024/04/07/letskerberos.html>

Her walk, in hindsight



Where do we go?



the Deployer

“The ethical stage: committing to an algorithm”

The whole menu

standard	what it is	status
ML-KEM (FIPS 203)	key exchange, lattices	shipping, everywhere
ML-DSA (FIPS 204)	signatures, lattices	the default
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- Everything else is either research prototype or in standardization process

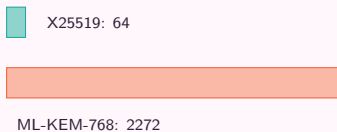
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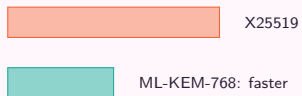
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- the question is *when* rather than *whether*.
- Betting against a decade of coordinated engineering effort is usually a bad bet.

“But it is bigger”

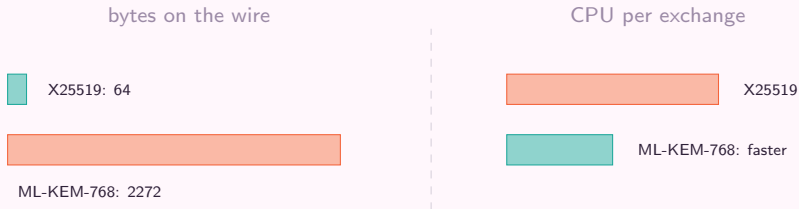
bytes on the wire



CPU per exchange

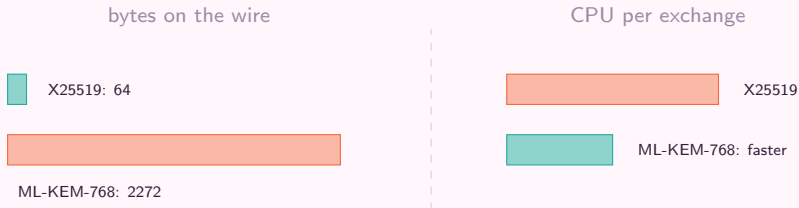


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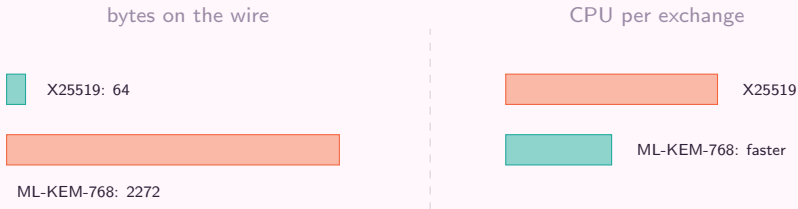
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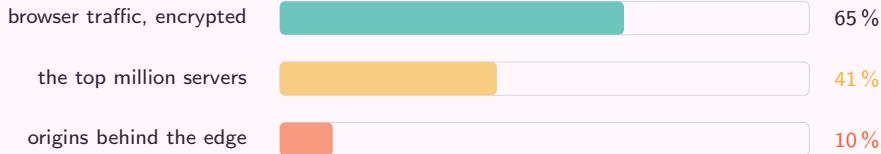
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- Though if you ship an Electron app, you have forfeited the right to complain about two kilobytes.

The state of the post-quantum internet

done



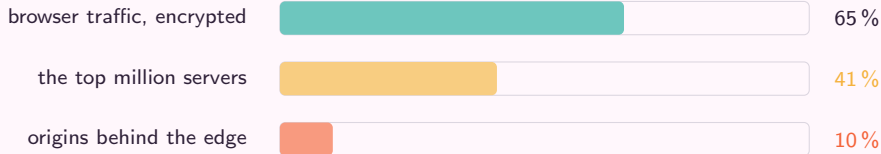
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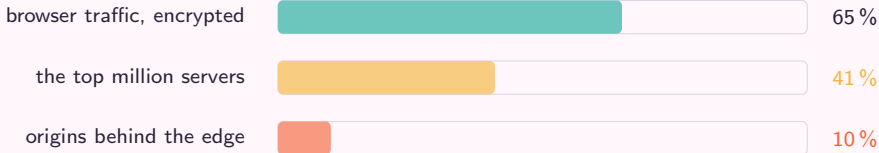
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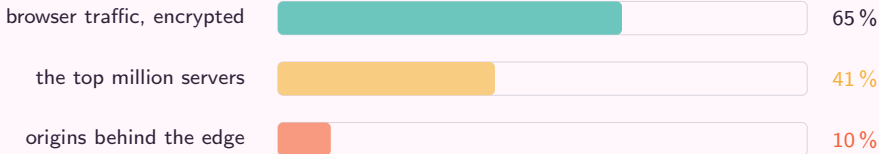
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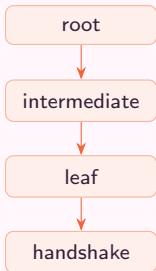
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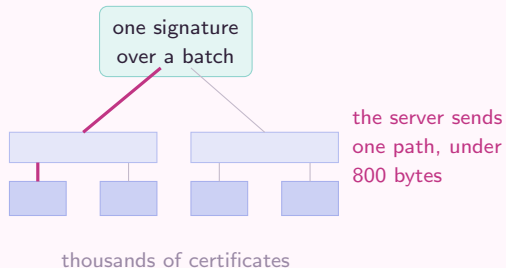
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- Cloudflare and Tranco scans, mid-2026.

If it can be broken, it can be fixed.

Merkle Tree Certificates

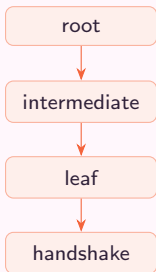


five signatures, sent every time

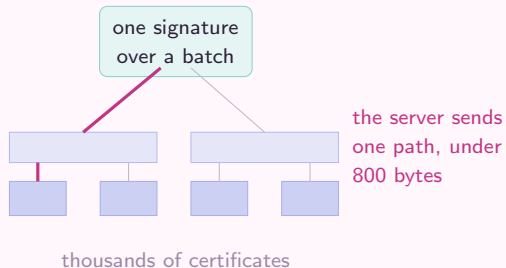


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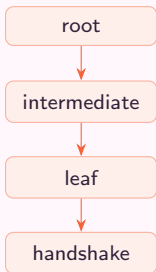


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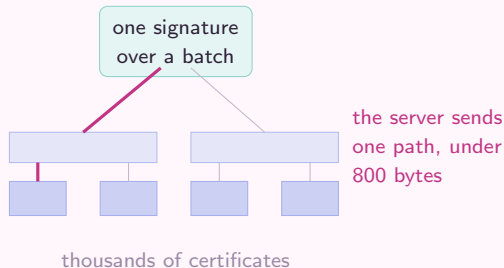


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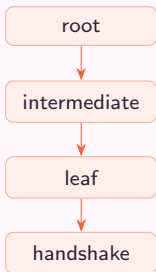


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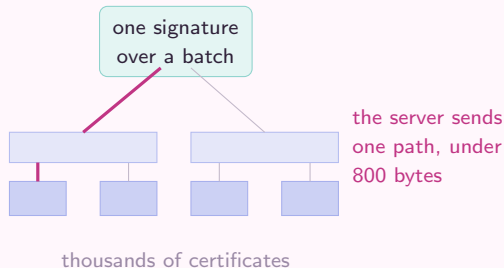


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- PQ signatures are large
- Sign batch once, proves membership: four of the five signatures disappear.
- trust moves from “verify a chain now” to “I already know the root”.
- In the common case this is smaller than what we send today, so post-quantum authentication could end up faster than classical

When does a useful quantum computer arrive?

- Estimates run from the early 2030s to the mid-2040s.
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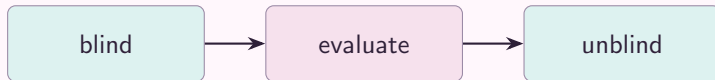
- Estimates run from the early 2030s to the mid-2040s.
- A lot of arguments: Hybrids, key sizes, which curve to keep
- Transition is as soon as 2029 (Google, Cloudflare)



the Academic

“What is to be done?”

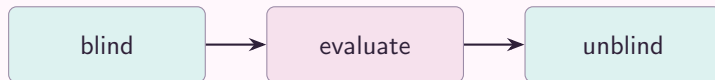
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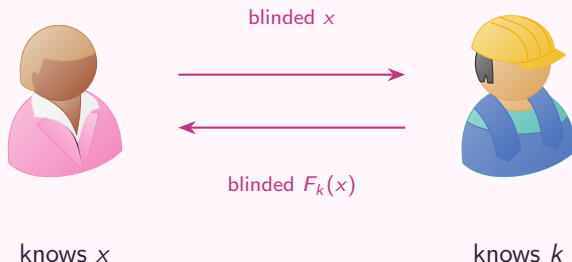
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- Need to slap on a lot of zero-knowledge proofs, counting arguments

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- **HAWK** fell to a classical attack.
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Actually strengthened lattices: conjecture that whole families of quantum lattice attacks may never gain more than a polynomial factor.
- Classic McEliece got a distinguisher, which became a quasi-polynomial key recovery. Nobody has run it, much like RSA-1024.

Building to break



- She learns $F_k(x)$. He learns nothing. Neither learns the other's secret.
- "Is my password in this breach?" without saying the password. Private contact discovery. Anonymous tokens.
- As an instrument: the smallest protocol that shows whether a primitive is **flexible**.

What now?



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I am afraid we will lose privacy on the way, because the tools that protect it are the ones that do not survive the move.

And still: the mathematics is beautiful, we get to open systems nobody has opened in thirty years, and almost none of it is decided yet.

Stop being sad.

Break something. On a laptop. Over a weekend.

- Sophie Schmieg's blog, especially keymaterial.net/2025/12/13/a-very-unscientific-guide-to-the-security-of-various-pqc-algorithms/
- Security Cryptography Whatever, especially <https://securitycryptographywhatever.com/2026/08/26/ai-lattice-proofs-with-chris-peikert/>
- Bas Westerbaan's blog, especially

Thank you.

heimberger.xyz